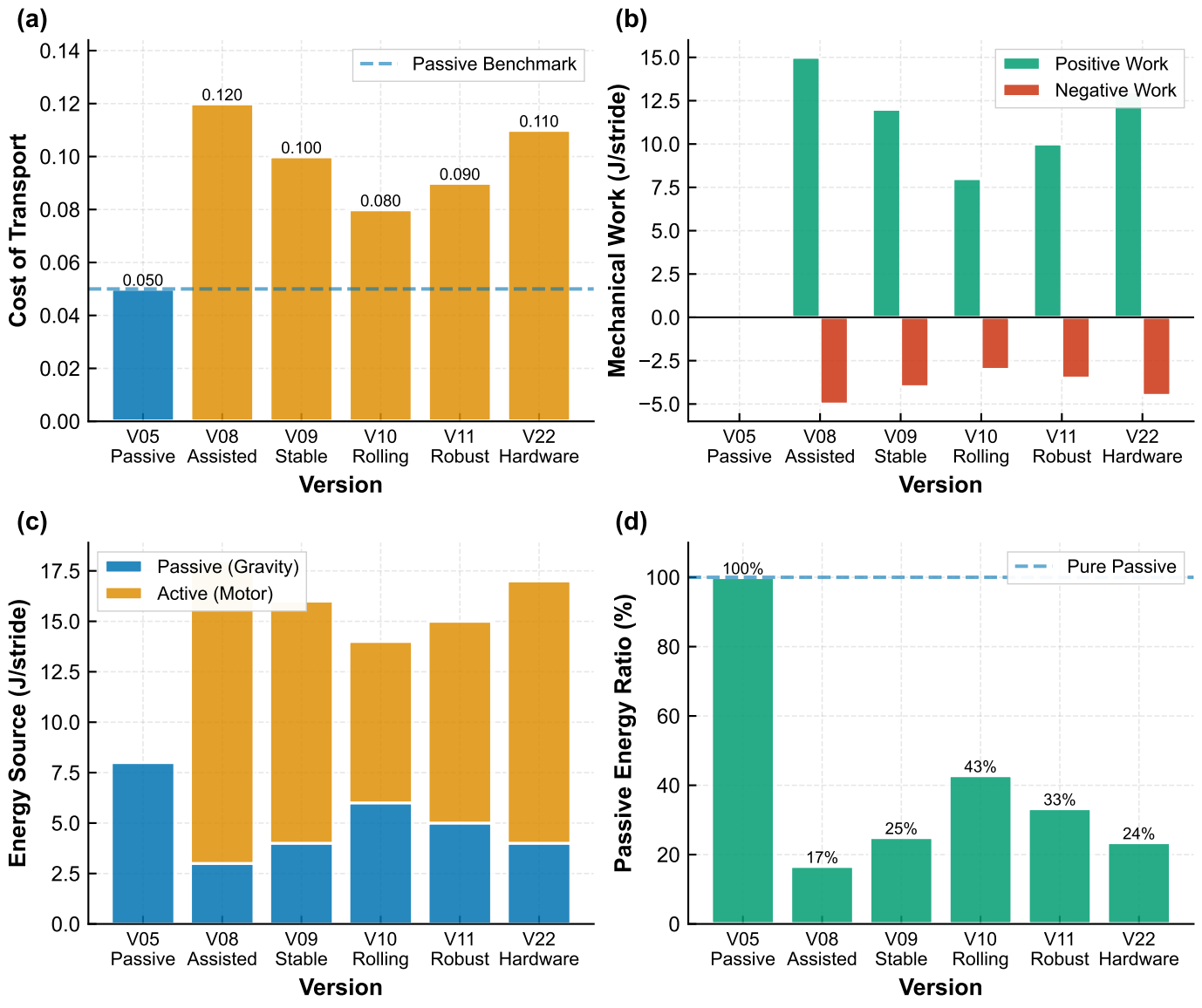
(a)
Cost of Transport
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Passive Benchmark
0.050
0.120
0.100
0.080
0.090
0.110
V05 Passive
V08 Assisted
V09 Stable
V10 Rolling
V11 Robust
V22 Hardware
Version
(b)
Mechanical Work (J/stride)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
Positive Work
Negative Work
V05 Passive
V08 Assisted
V09 Stable
V10 Rolling
V11 Robust
V22 Hardware
Version
(c)
Energy Source (J/stride)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Passive (Gravity)
Active (Motor)
V05 Passive
V08 Assisted
V09 Stable
V10 Rolling
V11 Robust
V22 Hardware
Version
(d)
Passive Energy Ratio (%)
100
80
60
40
20
0
Pure Passive
100%
17%
25%
43%
33%
24%
V05 Passive
V08 Assisted
V09 Stable
V10 Rolling
V11 Robust
V22 Hardware
Version